

A CONDITIONAL TWO-LEVEL REDUCTION FOR THE REAL-EXPONENT HUNTER MOMENT CONJECTURE

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STATUS

This is a conditional reduction, not a full solution. It proves that for real exponents $q > 2$, every positive-support constrained extremizer has at most two positive coefficient levels. Therefore Conjecture 4.8 of Brazitikos–Pandis is reduced, for $q > 5$, to a two-level Gamma inequality stated below. The earlier lane-5 packet already proves the conjecture for $0 < q \leq 5$.

SOURCE PROBLEM

The source paper is Silouanos Brazitikos and Christos Pandis, *Sharp inequalities for symmetric polynomials, Hunter’s conjecture, and moments of exponential random variables*, arXiv:2512.12254. Its Conjecture 4.8 asks: if $q > 0$, $a_i \geq 0$, $\sum_i a_i^2 = 1$, and X_i are independent standard exponential random variables, must

$$\mathbb{E}(a_1 X_1 + \cdots + a_n X_n)^q \geq \min_{1 \leq m \leq n} \rho(m, q), \quad \rho(m, q) = \mathbb{E} \left(\frac{X_1 + \cdots + X_m}{\sqrt{m}} \right)^q ?$$

We also conjecture that this behavior holds for every real $q > 0$, as expressed in the following conjecture:

Conjecture 4.8. Let $q > 0$ be a real number, and let $a_1, \dots, a_n$ be non-negative real numbers satisfying $\sum_{i=1}^n a_i^2 = 1$. If $X_1, \dots, X_n$ are i.i.d. standard exponential random variables, then

$$\mathbb{E}(a_1 X_1 + \cdots + a_n X_n)^q \geq \min\{\rho(1, q), \dots, \rho(n, q)\}.$$

4.1. Characterization of Extrema. We first introduce a Lemma that will be useful for the characterization of the global extrema

IDEA OF THE PROOF

For non-integer q , the moment is no longer a complete homogeneous symmetric polynomial, but it is still a divided difference. More precisely, after dividing by $\Gamma(q+1)$, the moment equals the divided difference of x^{q+n-1} at the coefficient nodes. The Lagrange multiplier equations therefore have the same structural consequence as in the integer proof: three distinct positive coefficient values would force two adjacent secant slopes of a divided-difference gradient to be equal. That is impossible because divided differences of x^{q+n-1} of the relevant order are strictly increasing in each node when $q > 2$.

Thus any remaining minimizer has only two coefficient levels. The full conjecture is then equivalent to checking a one-variable family

$$\frac{\mathbb{E}(xG_r + G_s)^q}{(rx^2 + s)^{q/2}},$$

where G_r, G_s are independent Gamma variables of integer shapes. The packet isolates this as the exact remaining dependency.

DIVIDED-DIFFERENCE PRELIMINARIES

For positive nodes $b_1, \dots, b_N$, including repeated nodes by the usual Hermite convention, write

$$[b_1, \dots, b_N]f$$

for the divided difference of order $N - 1$. We use two standard facts:

- (1) If $f \in C^{N-1}$ and the nodes are positive, then repeated-node divided differences are continuous limits of ordinary divided differences.
- (2) If $f^{(N)} > 0$ on $(0, \infty)$, then the divided difference $[b_1, \dots, b_N]f$ is strictly increasing in each node, unless the corresponding variation is trivial. This follows, for example, from the Hermite–Genocchi formula.

For $q > 0$ and $a_i > 0$, set

$$H_{n,q}(a_1, \dots, a_n) = \frac{1}{\Gamma(q+1)} \mathbb{E}(a_1 X_1 + \dots + a_n X_n)^q.$$

The hypoexponential moment formula gives

$$H_{n,q}(a_1, \dots, a_n) = [a_1, \dots, a_n] f_{n,q}, \quad f_{n,q}(x) = x^{q+n-1}.$$

Equivalently, this follows first for distinct a_i by partial fractions and then for repeated nodes by continuity.

EXTREMIZERS HAVE AT MOST TWO LEVELS

Lemma 1. *Let $q > 2$. Fix a positive coordinate support of size n . Every relative-interior local extremum of $H_{n,q}$ under $\sum_{i=1}^n a_i^2 = 1$ has at most two distinct coefficient values.*

Proof. Let $f = f_{n,q}$. The derivative of a divided difference with respect to a node is the divided difference with that node repeated:

$$\partial_i H_{n,q}(a) = [a_1, \dots, a_i, a_i, \dots, a_n]f.$$

For $i \neq j$, subtracting these two repeated-node divided differences gives

$$(1) \quad \frac{\partial_i H_{n,q}(a) - \partial_j H_{n,q}(a)}{a_i - a_j} = [a_1, \dots, a_n, a_i, a_j]f.$$

This identity is first checked for distinct nodes from the Newton divided difference recursion and then extended to repeated nodes by continuity.

At a relative-interior constrained extremum there is a Lagrange multiplier λ such that

$$\partial_i H_{n,q}(a) = 2\lambda a_i, \quad i = 1, \dots, n.$$

Hence the quotient in (1) is equal to 2λ for every pair of distinct coefficient values.

Suppose, toward a contradiction, that three positive coefficient values $x < y < z$ occur. Let $Q_{uv} = [a_1, \dots, a_n, u, v]f$. The multiset defining Q_{yz} is obtained from the multiset defining Q_{xy}

by replacing one extra copy of x with one extra copy of z , leaving all other nodes the same after sorting. Since

$$f^{(n+2)}(t) = (q+n-1)(q+n-2)\cdots(q-2)t^{q-3} > 0$$

for $t > 0$, divided-difference monotonicity gives $Q_{yz} > Q_{xy}$. But the Lagrange equations give $Q_{yz} = 2\lambda = Q_{xy}$, a contradiction. Thus at most two positive coefficient values can occur. $\square$

CONDITIONAL DEPENDENCIES

The remaining dependency is the following scalar two-level inequality.

Lemma 2 (Two-level Gamma inequality, unproved). *Let $q > 5$, let $r, s \geq 1$ be integers, and let G_r, G_s be independent Gamma random variables with shapes r, s and unit scale. Then for every $x \geq 0$,*

$$\frac{\mathbb{E}(xG_r + G_s)^q}{(rx^2 + s)^{q/2}} \geq \min_{1 \leq m \leq r+s} \rho(m, q).$$

This packet does not prove Lemma 2. The numerical checks in the code folder found no violation, but those checks are not proof.

Theorem 1 (Conditional completion of Conjecture 4.8 for $q > 5$). *Assume Lemma 2. Then Conjecture 4.8 of arXiv:2512.12254 holds for every real $q > 5$. Together with the earlier lane-5 $0 < q \leq 5$ packet, this would prove the conjecture for all $q > 0$.*

Proof. Fix $q > 5$, n , and $a_i \geq 0$ with $\sum_i a_i^2 = 1$. By compactness, the moment has a global minimizer on the positive orthant of the sphere. Remove zero coordinates and work on its positive support, of size $d \leq n$. By the preceding lemma, the positive coefficients of the minimizer have at most two distinct values. If there is only one value, the minimizer is an equal-support vector and gives exactly $\rho(d, q)$, which is at least $\min_{1 \leq m \leq n} \rho(m, q)$.

If there are two values, write them as α repeated r times and β repeated s times, with $r + s = d$. If $\beta > 0$, set $x = \alpha/\beta$. Since $r\alpha^2 + s\beta^2 = 1$,

$$\mathbb{E}(\alpha G_r + \beta G_s)^q = \frac{\mathbb{E}(xG_r + G_s)^q}{(rx^2 + s)^{q/2}}.$$

Lemma 2 gives this quantity at least $\min_{1 \leq m \leq d} \rho(m, q)$, and this is at least the weaker ambient lower bound $\min_{1 \leq m \leq n} \rho(m, q)$. The cases $x = 0$ or $\beta = 0$ are boundary equal-support cases and are included by continuity. This proves the conditional claim. $\square$

VERIFICATION NOTES

The code folder contains two non-proof checks.

- `pair_splitting_scan.py` checks exact Gamma–Dirichlet quadrature slices with one or two background coefficient blocks plus a split pair. In the recorded runs, no off-center pair-splitting minima were found.
- `exact_vector_search.py` screens arbitrary vectors using the exact hypoexponential divided-difference formula and validates the best candidates with high-precision arithmetic. The validated candidates found in the run collapsed back to equal supports or stayed above the conjectured threshold.

These checks support the two-level dependency but do not establish it.

NOVELTY AND LIMITATIONS

Local run indexes on 2026-07-18 showed the previous lane-5 $0 < q \leq 5$ partial packet and a serious full-push attempt for arXiv:2512.12254, but no packet recording this divided-difference two-level reduction. The adjacent later paper arXiv:2602.03058, already recorded in the earlier packet, covers the related Schur-concavity range $0 \leq q \leq 4$, not this conditional reduction.

No fresh external web search was performed for this conditional packet. The novelty confidence is therefore local-run confidence, not a literature-wide claim.

REFERENCES

REFERENCES

- [1] S. Brazitikos and C. Pandis, *Sharp inequalities for symmetric polynomials, Hunter's conjecture, and moments of exponential random variables*, arXiv:2512.12254.
- [2] S. Brazitikos, A. Giannopoulos, P. Valettas and B.-H. Vritsiou, *Geometry of isotropic convex bodies*, Mathematical Surveys and Monographs, vol. 196, American Mathematical Society, 2014.

HUMAN REVIEW RECOMMENDATION

Review as a likely valid conditional reduction. The main proof check is the divided-difference monotonicity step in the extremizer lemma. The main open task is to prove Lemma 2, or to find a counterexample to it.